Geometric sequences

1 For each of the following sequences:

i Identify if the sequence is arithmetic or geometric.

ii Write the common ratio or difference.

a $11, -99, 891, -8019, \ldots$ **b** $2, 6, 10, 14, \ldots$

2 Determine whether each of the following is a geometric progression:

a $4, -4, 4, -4, \ldots$ **b** $1, \sqrt{6}, 6, 6\sqrt{6}, \ldots$ **c** $2, 0, -2, -4, \ldots$ **d** $2, 2^2, 2^4, 2^6, \ldots$

3 Suppose $u_1, u_2, u_3, u_4, u_5, \ldots$ is a geometric sequence.
Is $u_1, u_3, u_5, \ldots$ a geometric sequence?

4 The first four terms of a geometric sequence are $-8, -16, -32, -64$. Evaluate:

a $\dfrac{u_2}{u_1}$ **b** $\dfrac{u_3}{u_2}$ **c** $\dfrac{u_4}{u_3}$ **d** u_5

5 Explain how the common ratio of a geometric sequence can be obtained.

6 Find the common ratio of the geometric sequence: $-70.4, -17.6, -4.4, -1.1, \ldots$

7 The first two terms of a geometric sequence are $\sqrt{5} + \sqrt{3}$ and $\sqrt{5} - \sqrt{3}$.
Find the common ratio.

8 For each of the following, write the first four terms in the geometric progression:

a The first term is 6 and the common ratio is 4.

b The first term is 7 and the common ratio is -2.

9 Write down the next two terms in the following sequences:

a $4, 12, 36$ **b** $12, -48, 192$ **c** $1, \dfrac{3}{4}, \dfrac{9}{16}$ **d** $-6, 9, -\dfrac{27}{2}$

10 Find the next two terms in the sequence in terms of n:

$$n - 5, n^2 + 5n, n^3 - 5n^2, n^4 + 5n^3, \ldots$$

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11 For each defined nth term of a sequence:

i State the first four terms.

ii Find the common ratio, r.

a $u_n = 3 \times 4^{n-1}$

b $u_n = -4 \times (-3)^{n-1}$

12 Use the common ratio to find the missing terms in the following geometric progressions:

a $-5, \square, -80, 320, \square$

b $\square, \square, \frac{3}{25}, -\frac{3}{125}, \square$

c $18, \square, \square, \square, \frac{32}{9}$

Technology

13 The 6th term of a geometric sequence is 557 and the 13th term is $255\,642$.

a Write an equation involving u_1, the first term, and r, the common ratio, for the 6th term.

b Write an equation involving u_1, the first term, and r, the common ratio, for the 13th term.

c Give the value of u_1 to the nearest integer and give the value of r to one decimal place. Assume u_1 and r are positive.

d Hence, determine the 8th term of the sequence.

14 The 7th term of a geometric sequence is 353 and the 13th term is 42.

a Write an equation involving u_1, the first term, and r, the common ratio, of this geometric sequence for the 7th term.

b Write an equation involving u_1, the first term, and r, the common ratio, of this geometric sequence for the 13th term.

c Use the simultaneous solving facility of your calculator to find the values of u_1 and r. Give the value of u_1 to the nearest integer and give the value of r to one decimal place. Assume u_1 and r are positive.

d Hence, determine the 9th term of the series.

15 The 5th term of a geometric sequence is 11 and the 12th term is 72.

a Write an equation involving u_1, the first term, and r, the common ratio, for the 5th term.

b Write an equation involving u_1, the first term, and r, the common ratio, for the 12th term.

c Use the simultaneous solving facility of your calculator to find the values of u_1 and r. Give the value of u_1 to the nearest integer and give the value of r to one decimal place. Assume u_1 and r are positive.

d Hence, state the general term, u_n, of the sequence.

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16 The 4th term of a geometric sequence is 33 and the 14th term is 952.

- **a** Write an equation involving u_1, the first term, and r, the common ratio, for the 4th term.
- **b** Write an equation involving u_1, the first term, and r, the common ratio, for the 14th term.
- **c** Give the value of u_1 to the nearest integer and give the value of r to one decimal place in the general term. Assume u_1 and r are positive.

17 The first three terms of a geometric sequence are $x + 11, x + 2$ and $x - 4$.

- **a** Form an equation in terms of x and then use a graphing calculator to find x.
- **b** Hence, find the common ratio of the sequence.
- **c** Hence, determine the 4th term of the sequence.

Applications

18 The following table shows the mass of 800 grams of radioactive element D left each day, given that element D loses half its mass every day.

- **a** Complete the table.
- **b** What type of decay is this, linear or exponential?

Day	Mass of element D (g)
0	800
1	
2	
3	
4	

19 The zoom function in a camera multiplies the dimensions of an image. In an image, the height of waterfall is 30 mm. After the zoom function is applied once, the height of the waterfall in the image is 36 mm. After a second application, its height is 43.2 mm.

- **a** Each time the zoom function is applied, by what factor is the image enlarged?
- **b** If the zoom function is applied a third time, find the exact height of the waterfall in the image.

20 Suppose you save $\$1$ the first day of a month, $\$2$ the second day, $\$4$ the third day, $\$8$ the fourth day, and so on. That is, each day you save twice as much as you did the day before.

- **a** How much will you save on the 17th day of the month?
- **b** How much will you save on the 29th day of the month?

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21 A car enthusiast purchases a vintage car for $\$220\,000$. Each year, its value increases at a rate of 12 percent of its value at the beginning of the year.
Find its value after 7 years, to two decimal places.

22 A gym trainer posts Monday's training program on the board, along with how you should progress each day that follows based on your level of fitness:

MONDAY TRAINING PROGRAM

Single rope skips	9
Weight lift	$6\frac{1}{2}$ kg
Rest	2 minutes
Row	$\frac{1}{4}$ mile

BEGINNER LEVEL:
Each day, increase the numbers and time by $\frac{1}{3}$ of the first day.
INTERMEDIATE LEVEL:
Each day, increase the numbers and time by $\frac{1}{3}$ of the previous day.

a Using the Intermediate Level training program:

i Find the number of single rope skips you would need to complete on Wednesday.

ii Find the weight you would need to weight lift on Wednesday as a mixed number.

iii Find the rest time on Wednesday.

iv Find the distance to be rowed on Wednesday.

b Using the Beginner Level training program:

i Find the number of single rope skips that will need to be done on Wednesday.

ii Find the distance you would need to row on Wednesday.

c Which level training plan is the most realistic in the long term, Beginner or Intermediate?

23 A sample of 2600 bacteria was taken to see how rapidly the bacteria would spread. After 1 day, the number of bacteria was found to be 2912.

a By what percentage had the number of bacteria increased over a period of one day?

b If the bacteria continue to multiply at this rate each day, what will the number of bacteria grow to eighteen days after the sample was taken? Round to the nearest whole number.

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24 A rectangular poster originally measures 81 centimetres in width and 256 centimetres in length. To edit the poster once, the length of the rectangle is decreased by $\frac{1}{4}$ and the width is increased by $\frac{1}{3}$.

a If the poster is edited once, find the ratio of the original area of the rectangle to the new area.

b If the edit is repeated 3 times, find the new area of the poster to the nearest square centimetre.

c Find the number of times, n, that the process must be repeated to produce a square poster.

25 To test the effectiveness of a new antibiotic, a certain bacteria is introduced to a body and the number of bacteria is monitored. Initially, there are 19 bacteria in the body, and after four hours, the number is found to double.

a If the bacterial population continues to double every four hours, how many bacteria will there be in the body after 24 hours?

b The antibiotic is applied after 24 hours, and is found to kill one third of the germs every two hours.
How many bacteria will there be left in the body 24 hours after applying the antibiotic? Assume the bacteria stops multiplying and round your answer to the nearest integer if necessary.

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